

ECEN 629 Convex Optimization for Electrical Engineering

Chapter 8. Geometric problems

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Minimum volume ellipsoid around a set

Löwner-John ellipsoid of a set C : minimum volume ellipsoid $\mathcal{E}$ s.t. $C \subseteq \mathcal{E}$

- parametrize $\mathcal{E}$ as $\mathcal{E} = \{v \mid \|Av + b\|_2 \leq 1\}$; w.l.o.g. assume $A \in \mathbf{S}_{++}^n$
- $\text{vol } \mathcal{E}$ is proportional to $\det A^{-1}$; to compute minimum volume ellipsoid,

$$\begin{array}{ll} \text{minimize (over } A, b) & \log \det A^{-1} \\ \text{subject to} & \sup_{v \in C} \|Av + b\|_2 \leq 1 \end{array}$$

convex, but evaluating the constraint can be hard (for general C)

finite set $C = \{x_1, \dots, x_m\}$:

$$\begin{array}{ll} \text{minimize (over } A, b) & \log \det A^{-1} \\ \text{subject to} & \|Ax_i + b\|_2 \leq 1, \quad i = 1, \dots, m \end{array}$$

also gives Löwner-John ellipsoid for polyhedron $\text{conv}\{x_1, \dots, x_m\}$

Maximum volume inscribed ellipsoid

maximum volume ellipsoid $\mathcal{E}$ inside a convex set $C \subseteq \mathbf{R}^n$

- parametrize $\mathcal{E}$ as $\mathcal{E} = \{Bu + d \mid \|u\|_2 \leq 1\}$; w.l.o.g. assume $B \in \mathbf{S}_{++}^n$
- $\text{vol } \mathcal{E}$ is proportional to $\det B$; can compute $\mathcal{E}$ by solving

$$\begin{array}{ll}\text{maximize} & \log \det B \\ \text{subject to} & \sup_{\|u\|_2 \leq 1} I_C(Bu + d) \leq 0\end{array}$$

(where $I_C(x) = 0$ for $x \in C$ and $I_C(x) = \infty$ for $x \notin C$)

convex, but evaluating the constraint can be hard (for general C)

polyhedron $\{x \mid a_i^T x \leq b_i, i = 1, \dots, m\}$:

$$\begin{array}{ll}\text{maximize} & \log \det B \\ \text{subject to} & \|Ba_i\|_2 + a_i^T d \leq b_i, \quad i = 1, \dots, m\end{array}$$

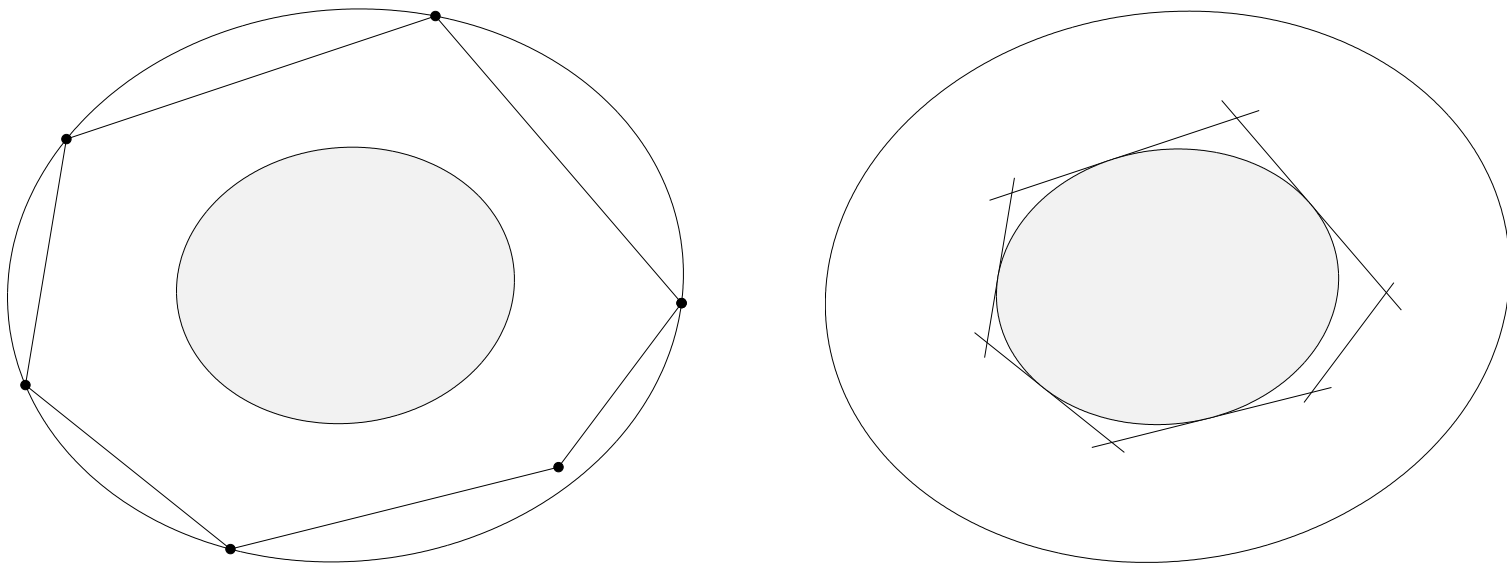
(constraint follows from $\sup_{\|u\|_2 \leq 1} a_i^T (Bu + d) = \|Ba_i\|_2 + a_i^T d$)

Efficiency of ellipsoidal approximations

$C \subseteq \mathbf{R}^n$ convex, bounded, with nonempty interior

- Löwner-John ellipsoid, shrunk by a factor n , lies inside C
- maximum volume inscribed ellipsoid, expanded by a factor n , covers C

example (for two polyhedra in $\mathbf{R}^2$)



factor n can be improved to $\sqrt{n}$ if C is symmetric

Exercise: Maximum volume rectangle inside a polyhedron

Show that the problem of finding the rectangle

$$\mathcal{R} = \{x \in \mathbf{R}^n \mid l \preceq x \preceq u\}$$

of the maximum volume, enclosed in a polyhedron

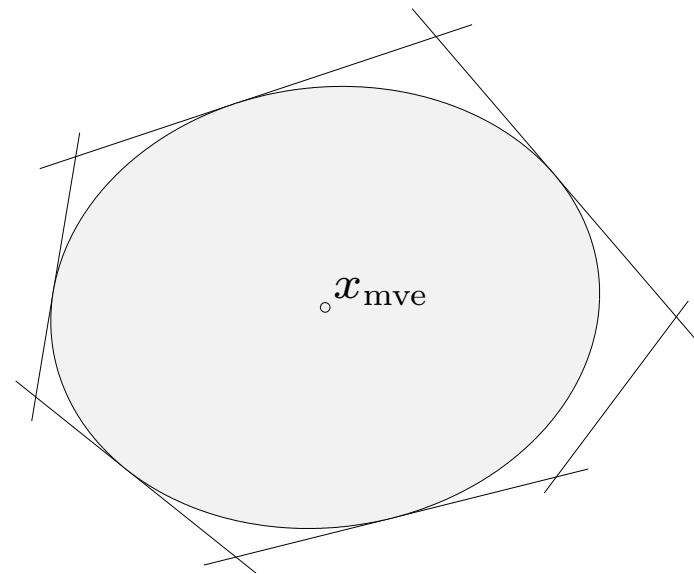
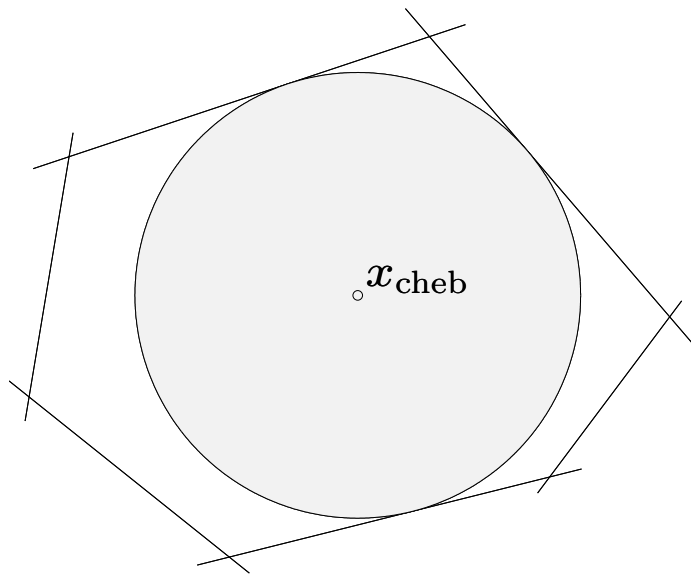
$$\mathcal{P} = \{x \in \mathbf{R}^n \mid Ax \preceq b\}$$

can be written as a convex optimization problem. (Hint: count the number of variables and constraints in the formulation)

Centering

some possible definitions of 'center' of a convex set C :

- center of largest inscribed ball ('Chebyshev center')
for polyhedron, can be computed via linear programming (page 4–19)
- center of maximum volume inscribed ellipsoid (page 8–3)



MVE center is invariant under affine coordinate transformations

Chebyshev center

Depth of a point $x \in C$

- The distance to the closest point in the exterior of C .
- The radius of the largest ball, centered at x , that lies in C .

$$\mathbf{depth}(x, C) = \mathbf{dist}(x, \mathbb{R}^n \setminus C)$$

Chebyshev center

Depth of a point $x \in C$

- The distance to the closest point in the exterior of C .
- The radius of the largest ball, centered at x , that lies in C .

$$\mathbf{depth}(x, C) = \mathbf{dist}(x, \mathbb{R}^n \setminus C)$$

Chebyshev center of the set C

- A point inside C that is farthest from the exterior of C ;

$$x_{\text{cheb}}(C) = \arg \max \mathbf{depth}(x, C) = \arg \max \mathbf{dist}(x, \mathbb{R}^n \setminus C).$$

Chebyshev center of a convex set

For a convex set:

$$C = \{x | f_i(x) \leq 0, i = 1, 2, \dots, m\},$$

where f_i 's are convex functions.

maximize: R

subject to: $g_i(x, R) \leq 0, \quad i = 1, 2, \dots, m.$

where

$$g_i(x, R) = \sup_{||u|| \leq 1} f_i(x + Ru).$$

- Convex optimization problem (g_i convex?);
- Evaluating g_i is in fact convex maximization problem, but solvable in certain special cases.

Exercise: Chebyshev center of a polyhedron

Suppose C is defined by a set of linear inequalities $a_i^T x \leq b_i$, $i = 1, 2, \dots, m$, and the norm $\|\cdot\|$ is ℓ_2 .

1. Simplify the function $g_i(x, R)$;
2. Write down the Chebyshev center problem;
3. What if the norm is not ℓ_2 ?

Analytic center of a set of inequalities

the analytic center of set of convex inequalities and linear equations

$$f_i(x) \leq 0, \quad i = 1, \dots, m, \quad Fx = g$$

is defined as the optimal point of

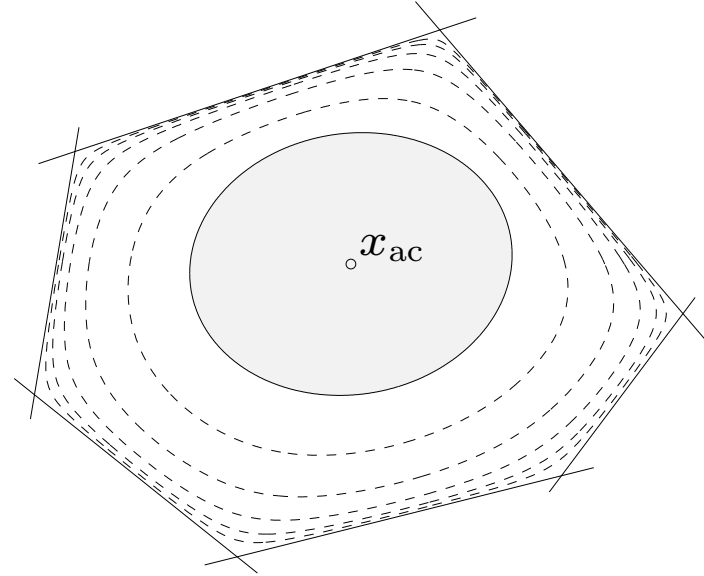
$$\begin{array}{ll} \text{minimize} & -\sum_{i=1}^m \log(-f_i(x)) \\ \text{subject to} & Fx = g \end{array}$$

- more easily computed than MVE or Chebyshev center (see later)
- not just a property of the feasible set: two sets of inequalities can describe the same set, but have different analytic centers

analytic center of linear inequalities $a_i^T x \leq b_i, i = 1, \dots, m$

x_{ac} is minimizer of

$$\phi(x) = - \sum_{i=1}^m \log(b_i - a_i^T x)$$



inner and outer ellipsoids from analytic center:

$$\mathcal{E}_{\text{inner}} \subseteq \{x \mid a_i^T x \leq b_i, i = 1, \dots, m\} \subseteq \mathcal{E}_{\text{outer}}$$

where

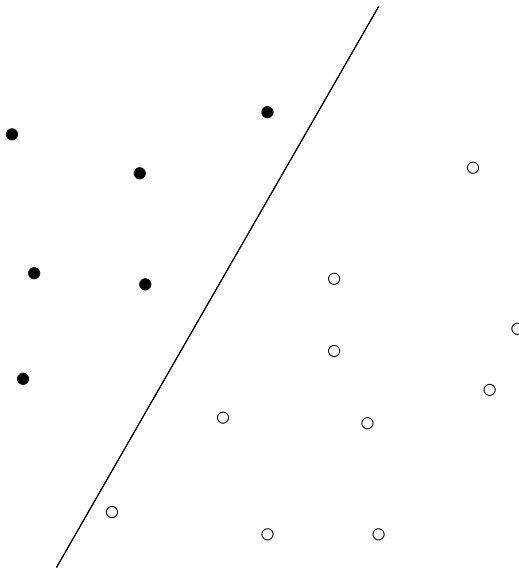
$$\mathcal{E}_{\text{inner}} = \{x \mid (x - x_{ac})^T \nabla^2 \phi(x_{ac})(x - x_{ac}) \leq 1\}$$

$$\mathcal{E}_{\text{outer}} = \{x \mid (x - x_{ac})^T \nabla^2 \phi(x_{ac})(x - x_{ac}) \leq m(m - 1)\}$$

Linear discrimination

separate two sets of points $\{x_1, \dots, x_N\}$, $\{y_1, \dots, y_M\}$ by a hyperplane:

$$a^T x_i + b > 0, \quad i = 1, \dots, N, \quad a^T y_i + b < 0, \quad i = 1, \dots, M$$



homogeneous in a , b , hence equivalent to

$$a^T x_i + b \geq 1, \quad i = 1, \dots, N, \quad a^T y_i + b \leq -1, \quad i = 1, \dots, M$$

a set of linear inequalities in a , b

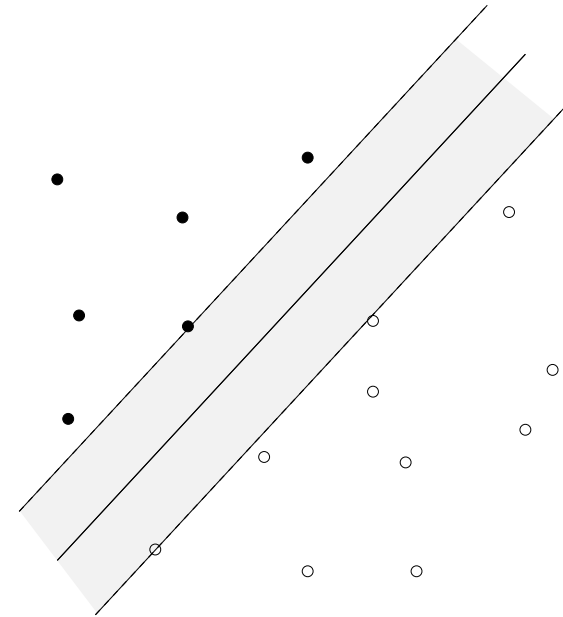
Robust linear discrimination

(Euclidean) distance between hyperplanes

$$\mathcal{H}_1 = \{z \mid a^T z + b = 1\}$$

$$\mathcal{H}_2 = \{z \mid a^T z + b = -1\}$$

is $\text{dist}(\mathcal{H}_1, \mathcal{H}_2) = 2/\|a\|_2$



to separate two sets of points by maximum margin,

$$\begin{aligned} &\text{minimize} && (1/2)\|a\|_2 \\ &\text{subject to} && a^T x_i + b \geq 1, \quad i = 1, \dots, N \\ & && a^T y_i + b \leq -1, \quad i = 1, \dots, M \end{aligned} \tag{1}$$

(after squaring objective) a QP in a, b

Exercise: Dual of Robust Linear Discrimination

The primal problem (squared objective function version) for the separable case

$$\begin{aligned} & \text{minimize: } \frac{1}{2} \|a\|_2^2 \\ & \text{subject to: } a^T x_i + b \geq 1, \quad i = 1, 2, \dots, N \\ & \quad \quad \quad a^T y_i + b \leq -1, \quad i = 1, 2, \dots, M. \end{aligned}$$

Derive the dual optimization problem.

Dual in the Separable case

The initial form

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^2 \\ \text{subject to: } & \sum \lambda_i = \sum \mu_i \\ & \lambda \succeq 0, \mu \succeq 0. \end{aligned}$$

Dual in the Separable case

The initial form

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^2 \\ \text{subject to: } & \sum \lambda_i = \sum \mu_i \\ & \lambda \succeq 0, \mu \succeq 0. \end{aligned}$$

Simplify it further: $c = \sum \lambda_i$, $\theta_i = \lambda_i/c$, $\gamma_i = \mu_i/c$ then maximize over c
 $\Rightarrow c^* = 2 \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^{-2}$
 $\Rightarrow$

$$\begin{aligned} \text{minimize: } & \left\| \sum_{i=1}^N \theta_i x_i - \sum_{i=1}^M \gamma_i y_i \right\|_2^2 \\ \text{subject to: } & \sum \theta_i = \sum \gamma_i = 1 \\ & \theta \succeq 0, \gamma \succeq 0. \end{aligned}$$

Lagrange dual of maximum margin separation problem (1)

$$\begin{aligned} & \text{maximize} && \mathbf{1}^T \lambda + \mathbf{1}^T \mu \\ & \text{subject to} && 2 \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2 \leq 1 \\ & && \mathbf{1}^T \lambda = \mathbf{1}^T \mu, \quad \lambda \succeq 0, \quad \mu \succeq 0 \end{aligned} \tag{2}$$

from duality, optimal value is inverse of maximum margin of separation

interpretation

- change variables to $\theta_i = \lambda_i / \mathbf{1}^T \lambda$, $\gamma_i = \mu_i / \mathbf{1}^T \mu$, $t = 1 / (\mathbf{1}^T \lambda + \mathbf{1}^T \mu)$
- invert objective to minimize $1 / (\mathbf{1}^T \lambda + \mathbf{1}^T \mu) = t$

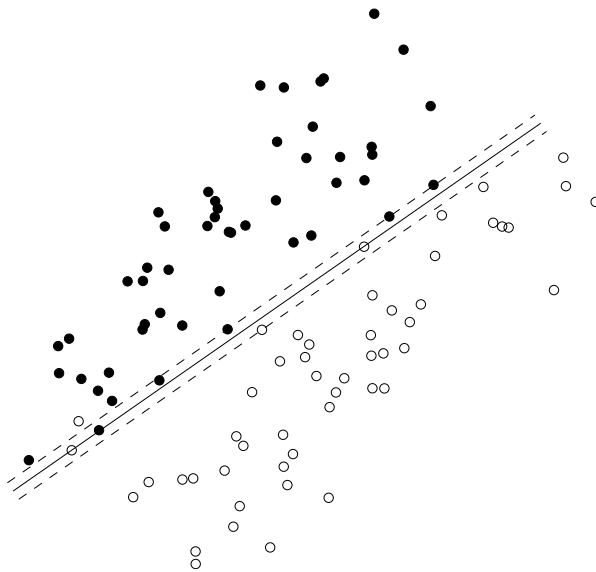
$$\begin{aligned} & \text{minimize} && t \\ & \text{subject to} && \left\| \sum_{i=1}^N \theta_i x_i - \sum_{i=1}^M \gamma_i y_i \right\|_2 \leq t \\ & && \theta \succeq 0, \quad \mathbf{1}^T \theta = 1, \quad \gamma \succeq 0, \quad \mathbf{1}^T \gamma = 1 \end{aligned}$$

optimal value is distance between convex hulls

Approximate linear separation of non-separable sets

$$\begin{array}{ll}\text{minimize} & \mathbf{1}^T u + \mathbf{1}^T v \\ \text{subject to} & a^T x_i + b \geq 1 - u_i, \quad i = 1, \dots, N \\ & a^T y_i + b \leq -1 + v_i, \quad i = 1, \dots, M \\ & u \succeq 0, \quad v \succeq 0\end{array}$$

- an LP in a, b, u, v
- at optimum, $u_i = \max\{0, 1 - a^T x_i - b\}$, $v_i = \max\{0, 1 + a^T y_i + b\}$
- can be interpreted as a heuristic for minimizing #misclassified points

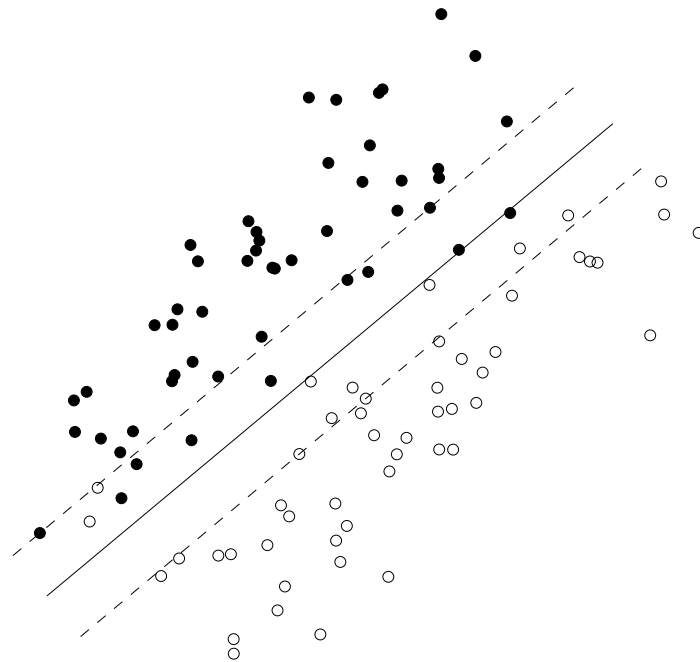


Support vector classifier

$$\begin{aligned} &\text{minimize} && \|a\|_2 + \gamma(\mathbf{1}^T u + \mathbf{1}^T v) \\ &\text{subject to} && a^T x_i + b \geq 1 - u_i, \quad i = 1, \dots, N \\ & && a^T y_i + b \leq -1 + v_i, \quad i = 1, \dots, M \\ & && u \succeq 0, \quad v \succeq 0 \end{aligned}$$

produces point on trade-off curve between inverse of margin $2/\|a\|_2$ and classification error, measured by total slack $\mathbf{1}^T u + \mathbf{1}^T v$

same example as previous page,
with $\gamma = 0.1$:



Support vector machine: Dual problem

The primal problem (again the squared norm version)

$$\begin{aligned} & \text{minimize: } \frac{1}{2} \|a\|_2^2 + \gamma(\mathbf{1}^T u + \mathbf{1}^T v) \\ & \text{subject to: } a^T x_i + b \geq 1 - u_i, \quad i = 1, \dots, N \\ & \quad \quad \quad a^T y_i + b \leq -1 + v_i, \quad i = 1, \dots, M \\ & \quad \quad \quad u \succeq 0, \quad v \succeq 0. \end{aligned}$$

Exercise: derive the dual problem.

Support vector machine: Dual problem

The primal problem (again the squared norm version)

$$\begin{aligned} \text{minimize: } & \frac{1}{2} \|a\|_2^2 + \gamma(\mathbf{1}^T u + \mathbf{1}^T v) \\ \text{subject to: } & a^T x_i + b \geq 1 - u_i, \quad i = 1, \dots, N \\ & a^T y_i + b \leq -1 + v_i, \quad i = 1, \dots, M \\ & u \succeq 0, \quad v \succeq 0. \end{aligned}$$

The dual problem

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^2 \\ \text{subject to: } & \sum_{i=1}^N \lambda_i = \sum_{i=1}^M \mu_i \\ & \gamma \succeq \lambda \succeq 0, \gamma \succeq \mu \succeq 0 \end{aligned}$$

Support vector machine: Separable vs. Non-separable

Non-separable (SVM):

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^2 \\ \text{subject to: } & \sum_{i=1}^N \lambda_i = \sum_{i=1}^M \mu_i \\ & \gamma \succeq \lambda \succeq 0, \gamma \succeq \mu \succeq 0 \end{aligned}$$

Separable (robust discrimination):

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \mu_i y_i \right\|_2^2 \\ \text{subject to: } & \sum_{i=1}^N \lambda_i = \sum_{i=1}^M \mu_i \\ & \lambda \succeq 0, \mu \succeq 0. \end{aligned}$$

Support vector machine: Dual problem

The dual problem

$$\begin{aligned} \text{maximize: } & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i - \frac{1}{2} \left\| \sum_{i=1}^N \lambda_i x_i - \sum_{i=1}^M \lambda_i y_i \right\|_2^2 \\ \text{subject to: } & \sum_{i=1}^N \lambda_i = \sum_{i=1}^M \mu_i \\ & \gamma \succeq \lambda \succeq 0, \gamma \succeq \mu \succeq 0 \end{aligned}$$

Sometimes the objective function is explicitly written as

$$\begin{aligned} & \sum_{i=1}^N \lambda_i + \sum_{i=1}^M \mu_i + \sum_{i=1}^N \sum_{j=1}^M \lambda_i \mu_j (x_i \cdot y_j) - \frac{1}{2} \sum_{i=1}^N \sum_{j=1}^N \lambda_i \lambda_j (x_i \cdot x_j) \\ & - \frac{1}{2} \sum_{i=1}^M \sum_{j=1}^M \mu_i \mu_j (y_i \cdot y_j) \end{aligned}$$

Support Vectors for the Non-separable Case

For the optimal solutions (complementary slackness)

$$\lambda_i = 0 \Rightarrow (a^T x_i + b) \geq 1$$

$$\mu_i = 0 \Rightarrow (a^T y_i + b) \leq -1$$

$$\lambda_i = \gamma \Rightarrow (a^T x_i + b) \leq 1$$

$$\mu_i = \gamma \Rightarrow (a^T y_i + b) \geq -1$$

$$0 < \lambda_i < \gamma \Rightarrow (a^T x_i + b) = 1$$

$$0 < \mu_i < \gamma \Rightarrow (a^T y_i + b) = -1$$

- The range $(0, \gamma)$ gives the support vectors, as in the separable case;
- The SVM also depends on the erroneously classified terms ($= \gamma$);
- How to recover the coefficients a and b ?

Nonlinear discrimination

separate two sets of points by a nonlinear function:

$$f(x_i) > 0, \quad i = 1, \dots, N, \quad f(y_i) < 0, \quad i = 1, \dots, M$$

- choose a linearly parametrized family of functions

$$f(z) = \theta^T F(z)$$

$F = (F_1, \dots, F_k) : \mathbf{R}^n \rightarrow \mathbf{R}^k$ are basis functions

- solve a set of linear inequalities in θ :

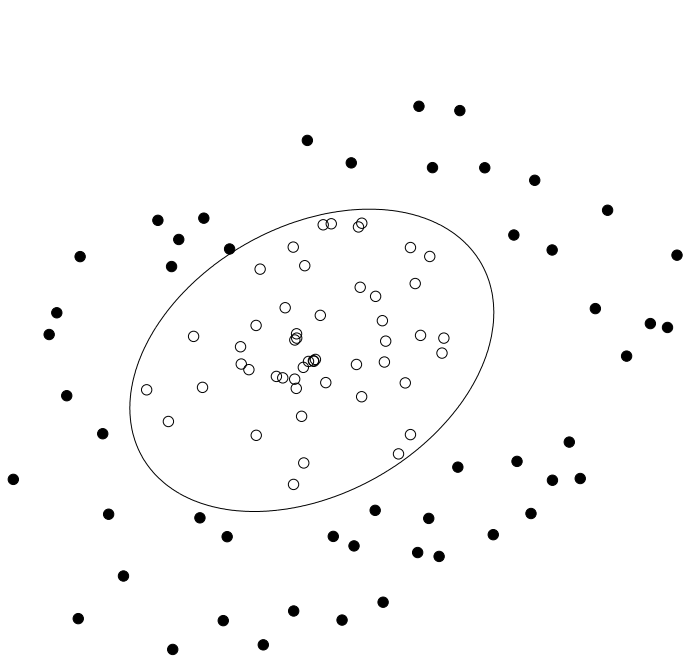
$$\theta^T F(x_i) \geq 1, \quad i = 1, \dots, N, \quad \theta^T F(y_i) \leq -1, \quad i = 1, \dots, M$$

quadratic discrimination: $f(z) = z^T P z + q^T z + r$

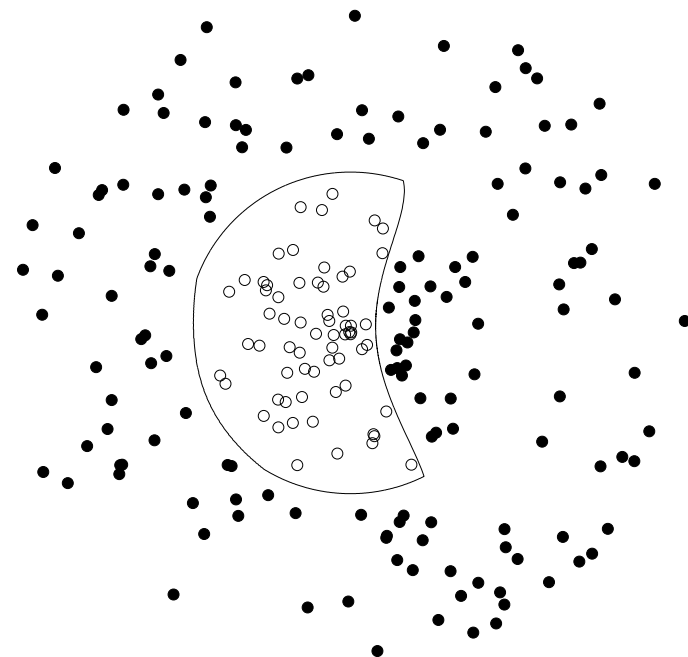
$$x_i^T P x_i + q^T x_i + r \geq 1, \quad y_i^T P y_i + q^T y_i + r \leq -1$$

can add additional constraints (*e.g.*, $P \preceq -I$ to separate by an ellipsoid)

polynomial discrimination: $F(z)$ are all monomials up to a given degree



separation by ellipsoid



separation by 4th degree polynomial

Primal vs. Dual

Why do we need to look at the dual

- Primal: dimension of a, b = dimension of the sample;
- Dual: dimension of λ, μ = number of samples;
- Which is smaller in practice? If the latter, more efficient to compute;
- Reveals more about the structure of the data;
- Numerically more stable to solve (though both QP).

Another important reason: the kernel trick

- The dual problem is effective only through the inner product of x_i and y_j ;
- If the discrimination is non-linear, but separable through some non-linear mapping: a mapping $\phi(x)$;
- Instead of computing $\phi(x_i)\phi(y_j)$ etc., simplification exists to compute it by computing $(x_i \cdot y_j)$.